

Big Data Analytics of Crime Patterns

using Hadoop MapReduce and Apache Hive

Your Group Members

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System Architecture

Pipeline Overview:

1. **Ingestion:** Dataset uploaded to HDFS
(/user/\$USER/bigdata_project/crimes)
2. **MapReduce:** Parallel processing of crime records to aggregate frequencies.
3. **Hive:** Data loaded into Hive via OpenCSVSerde for complex SQL analytics.

[Insert Futuristic Architecture Diagram Here]

MapReduce: Crime Type Counting

Goal: Calculate frequency of each primary crime type.

Mapper Logic:

- ▶ Parses each CSV line, handling quotes safely.
- ▶ Emits (PrimaryType, 1).

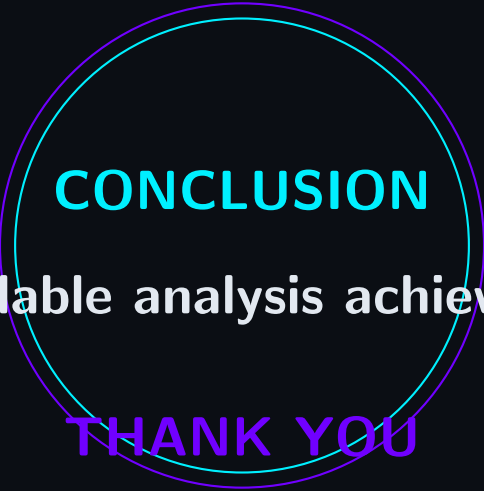
Reducer Logic:

- ▶ Iterates over values for each key.
- ▶ Emits (PrimaryType, TotalCount).

Apache Hive Analytics

Successfully implemented all required SQL clauses for deep insights:

- ▶ **SELECT & WHERE:** Filtering narcotic arrests.
- ▶ **ORDER BY:** Chronological listing of homicides.
- ▶ **GROUP BY & COUNT:** Crimes per location description.
- ▶ **HAVING:** Filtering crime types > 500 occurrences.
- ▶ **SUM & AVG:** Aggregating total arrests and average domestic crimes.
- ▶ **MAX & MIN:** Finding safest and most dangerous districts.
- ▶ **JOIN:** Displaying actual community names instead of just codes.



CONCLUSION

Scalable analysis achieved.

THANK YOU